

EDA Financial Discussions - Analysis Report

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Executive Summary

This report summarizes the exploratory data analysis conducted to identify suitable datasets for predicting engagement and sentiment surges in stock-related social media discussions.

Key Findings

- **Datasets discovered:** 0 (0 complete with engagement + sentiment fields)
- **API platforms assessed:** 2
- **Quality reports generated:** 7 (6 suitable)
- **Surge definitions evaluated:** 0 (0 viable with $\geq 2\%$ positive class)

Dataset Discovery Results

No datasets were discovered during the scan.

API Feasibility Findings

Twitter API

- **Historical access:** No
- **Supports surge label construction:** Yes
- **Estimated collection time:** 0.4 hours
- **Estimated cost:** \$100.00
- **Endpoints available:** 5

Rate Limits

- free_tier: {'tweets_per_month': 1500, 'requests_per_15min': 15, 'posts_per_request': 100}
- basic_tier: {'tweets_per_month': 10000, 'requests_per_15min': 60, 'posts_per_request': 100}
- pro_tier: {'tweets_per_month': 1000000, 'requests_per_15min': 300, 'posts_per_request': 100}

Cost Tiers

Tier	Details
Free	cost_usd_monthly: 0, tweet_cap: 1500
Basic	cost_usd_monthly: 100, tweet_cap: 10000
Pro	cost_usd_monthly: 5000, tweet_cap: 1000000
Enterprise	cost_usd_monthly: 42000, tweet_cap: 50000000

Paid Fields

The following fields require paid access:

- **impression_count**: requires Basic (\$100/month)
- **full archive search**: requires Pro (\$5,000/month)
- **quote_count**: requires Basic (\$100/month)

Reddit API

- **Historical access**: Yes
- **Supports surge label construction**: Yes
- **Estimated collection time**: 0.0 hours
- **Estimated cost**: \$0.00
- **Endpoints available**: 7

Rate Limits

- `oauth_tier`: {'requests_per_minute': 100, 'posts_per_request': 100, 'daily_limit': None}
- `free_tier_note`: Reddit API is free for non-commercial use with OAuth. Commercial use requires paid access.

Cost Tiers

Tier	Details
Free (non-commercial)	<code>cost_usd_monthly</code> : 0, <code>rate_limit</code> : 100 requests/min, note: Requires OAuth app registration
Commercial	<code>cost_usd_monthly</code> : Contact Reddit, <code>rate_limit</code> : Higher limits available, note: Required for commercial data use since 2023 API changes

Paid Fields

The following fields require paid access:

- **full historical archive**: requires Commercial (contact Reddit)
- **real-time streaming**: requires Commercial (contact Reddit)

EDA Statistics

StockMarket_subreddit.csv

Dataset Structure

- **Records**: 72,620
- **Tickers**: 1596
- **Columns**: 7
- **Date range**: 1970-01-01 to 1970-01-01
- **Recommendation**: suitable

Missing Values

Column	Missing %
selftext	29.2%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	6.3	1.0	7.0	16.0	97.0
num_comments	8.1	1.0	17.0	32.0	124.0

Sentiment Statistics

- **mean:** 0.092
- **median:** 0.000
- **std:** 0.305
- **Bullish/Bearish ratio:** 2.48

Identified Risks

- Duplicate rows detected: 1103 duplicates (1.5%), which may inflate engagement statistics.

submissions_reddit_1.csv

Dataset Structure

- **Records:** 775,326
- **Tickers:** 4450
- **Columns:** 23
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): selftext

Column	Missing %
selftext	33.8% 
link_flair_text	1.7%
title	0.0%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	116.0	1.0	21.0	70.0	964.0

Metric	Mean	Median	P90	P95	P99
num_comments	24.6	1.0	11.0	29.0	164.0

Sentiment Statistics

- **mean:** 0.070
- **median:** 0.000
- **std:** 0.340
- **Bullish/Bearish ratio:** 1.80

Identified Risks

- High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

submissions_reddit_2.csv

Dataset Structure

- **Records:** 75,857
- **Tickers:** 1962
- **Columns:** 23
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): link_flair_text

Column	Missing %
link_flair_text	46.9% 
selftext	0.1%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	30.2	1.0	14.0	39.0	413.0
num_comments	14.8	1.0	26.0	51.0	250.4

Sentiment Statistics

- **mean:** 0.085
- **median:** 0.000
- **std:** 0.282
- **Bullish/Bearish ratio:** 2.53

Identified Risks

- High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

submissions_reddit_3.csv

Dataset Structure

- **Records:** 43,809
- **Tickers:** 1474
- **Columns:** 23
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): selftext

Column	Missing %
selftext	39.1% ⚠
link_flair_text	13.8%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	35.0	1.0	18.0	65.0	848.9
num_comments	7.0	0.0	13.0	28.0	129.0

Sentiment Statistics

- **mean:** 0.104
- **median:** 0.000
- **std:** 0.318
- **Bullish/Bearish ratio:** 2.59

Identified Risks

- High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

submissions_reddit_4.csv

Dataset Structure

- **Records:** 131,181

- **Tickers:** 116
- **Columns:** 23
- **Date range:** 2021-01-24 to 2021-12-31
- **Recommendation:** suitable

Missing Values

Column	Missing %
link_flair_text	4.1%
selftext	0.5%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	10.1	1.0	5.0	9.0	138.0
num_comments	10.2	6.0	19.0	27.0	94.0

Sentiment Statistics

- **mean:** 0.071
- **median:** 0.000
- **std:** 0.312
- **Bullish/Bearish ratio:** 1.66

submissions_reddit_5.csv

Dataset Structure

- **Records:** 54,785
- **Tickers:** 2911
- **Columns:** 23
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

Column	Missing %
selftext	20.7%
link_flair_text	1.0%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	29.4	1.0	21.0	56.8	437.5
num_comments	10.8	1.0	21.0	39.0	137.0

Sentiment Statistics

- **mean:** 0.118
- **median:** 0.000
- **std:** 0.298
- **Bullish/Bearish ratio:** 3.50

train_data.csv

Dataset Structure

- **Records:** 16,990
- **Tickers:** 0
- **Columns:** 1
- **Date range:** unknown to unknown
- **Recommendation:** unsuitable

Missing Values

Column	Missing %
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Identified Risks

- Duplicate rows detected: 16970 duplicates (99.9%), which may inflate engagement statistics.
- No date column 'date' found: temporal analysis and 24-hour surge windowing cannot be performed.
- No ticker column 'ticker' found and ticker extraction from text yielded no results: per-ticker engagement normalization cannot be performed.

Failing EDA Objectives

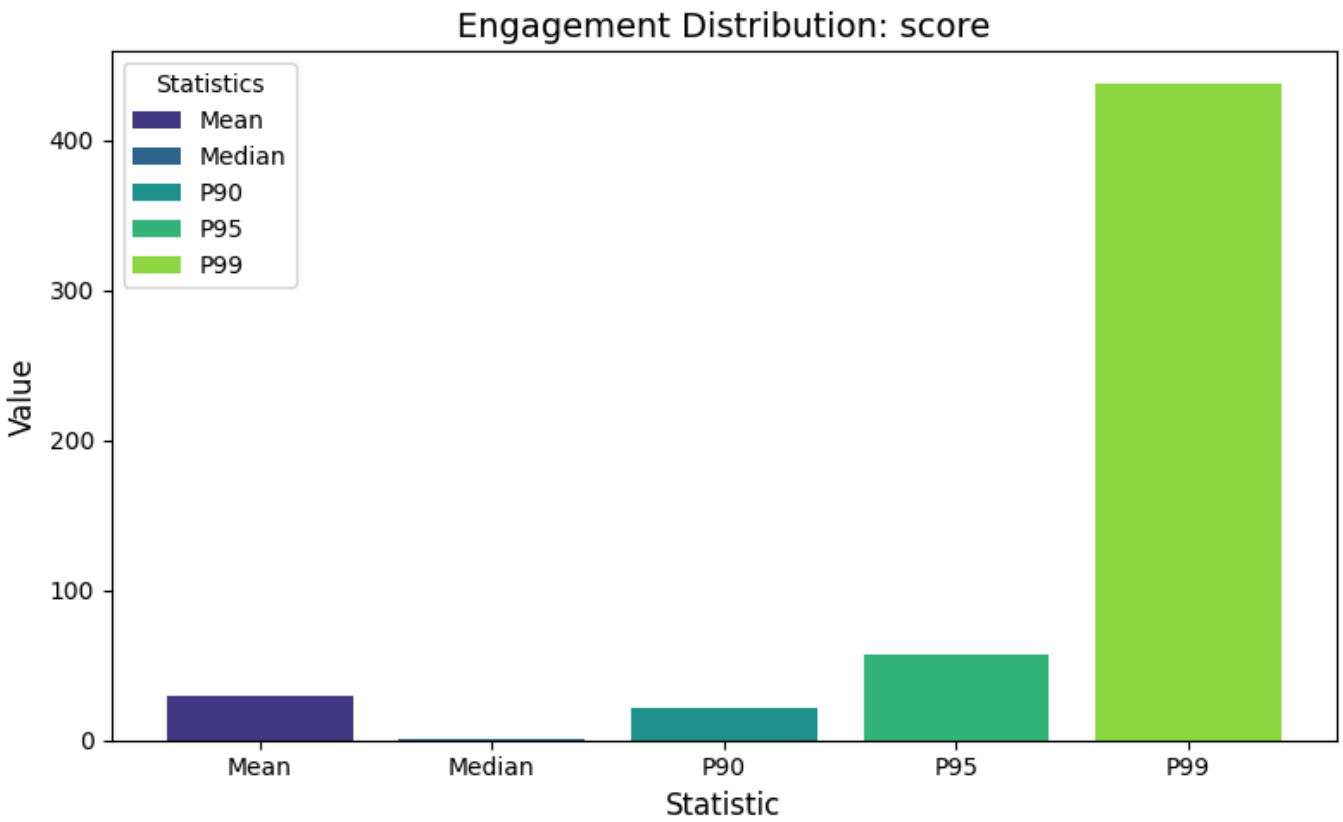
-  time_coverage
-  engagement_distributions
-  sentiment_distributions
-  surge_definitions

Surge Analysis Results

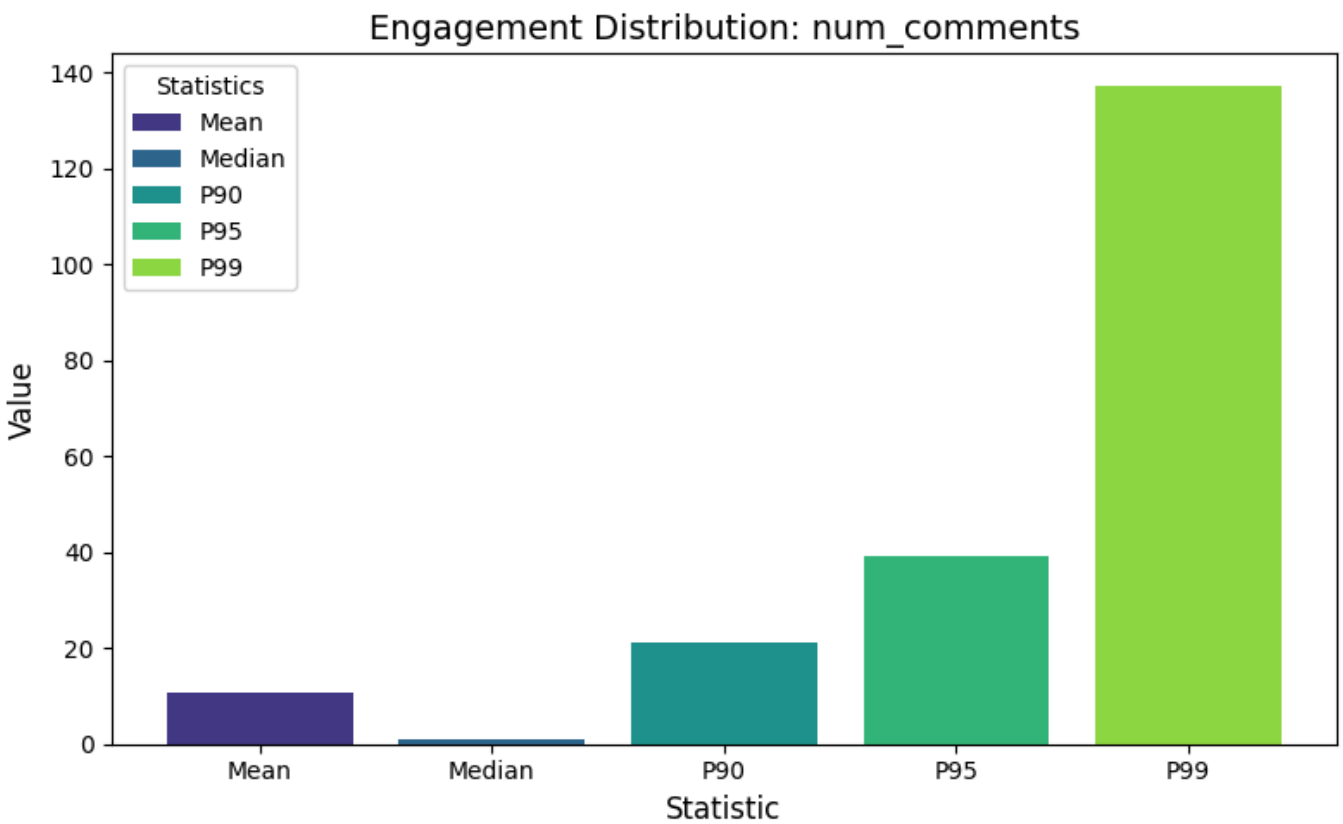
No surge analysis was performed.

Visualizations

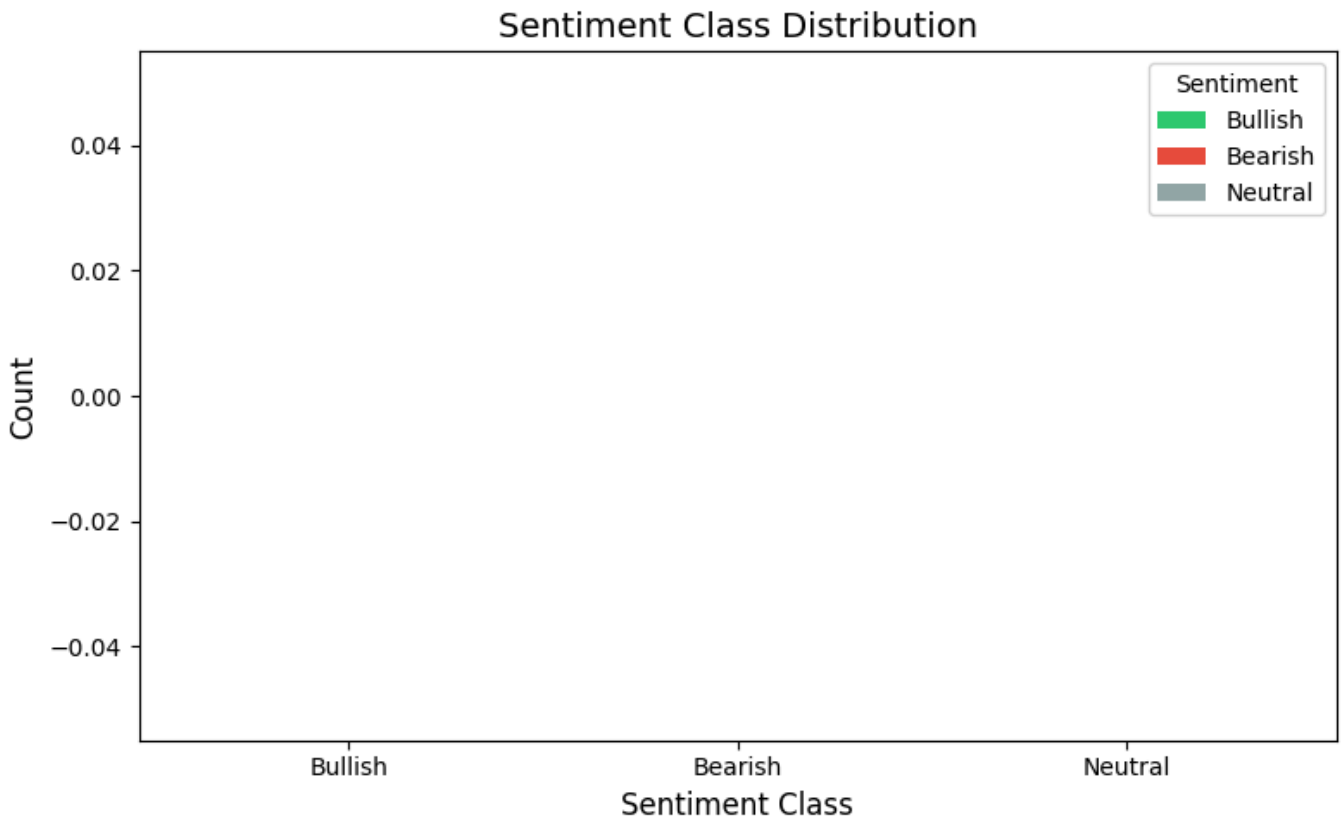
Engagement Distribution Score



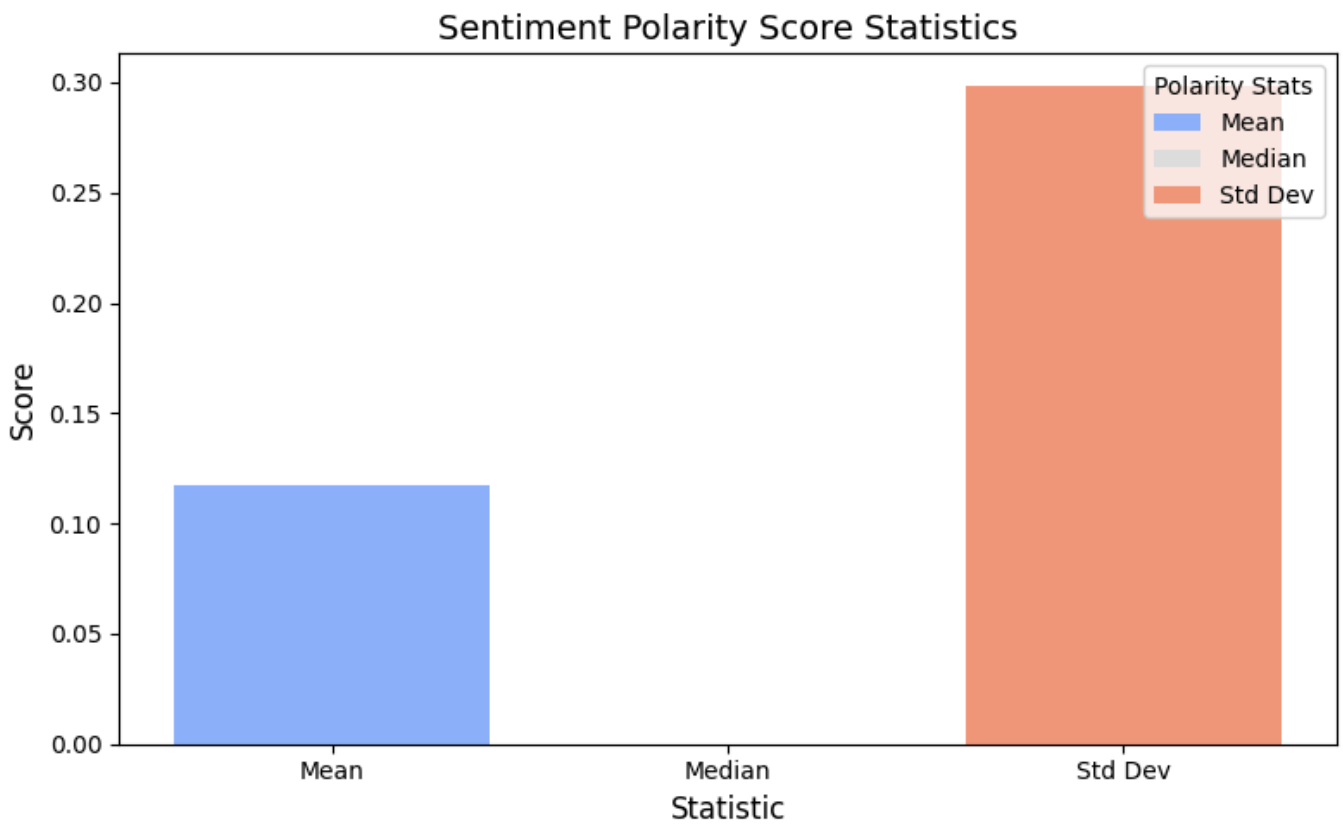
Engagement Distribution Num Comments



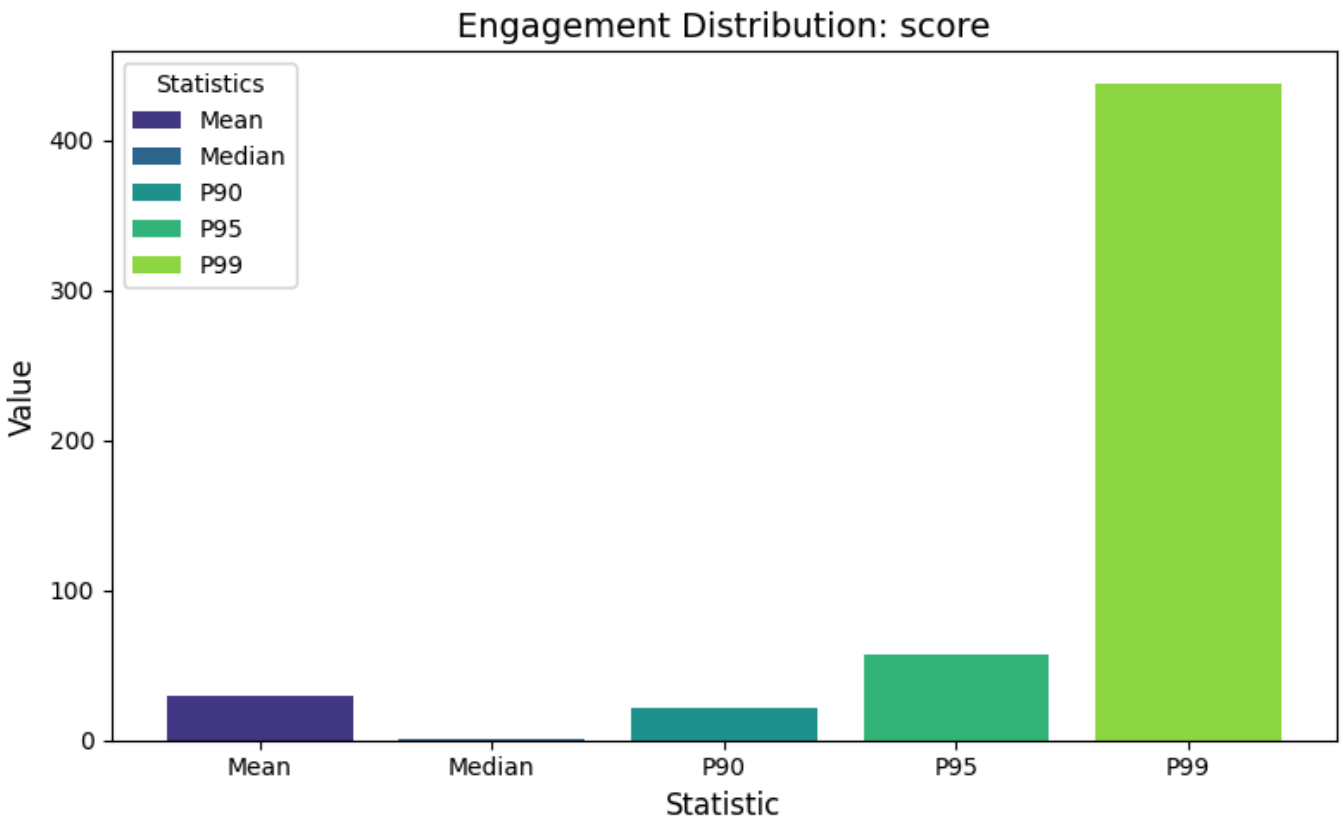
Sentiment Class Distribution



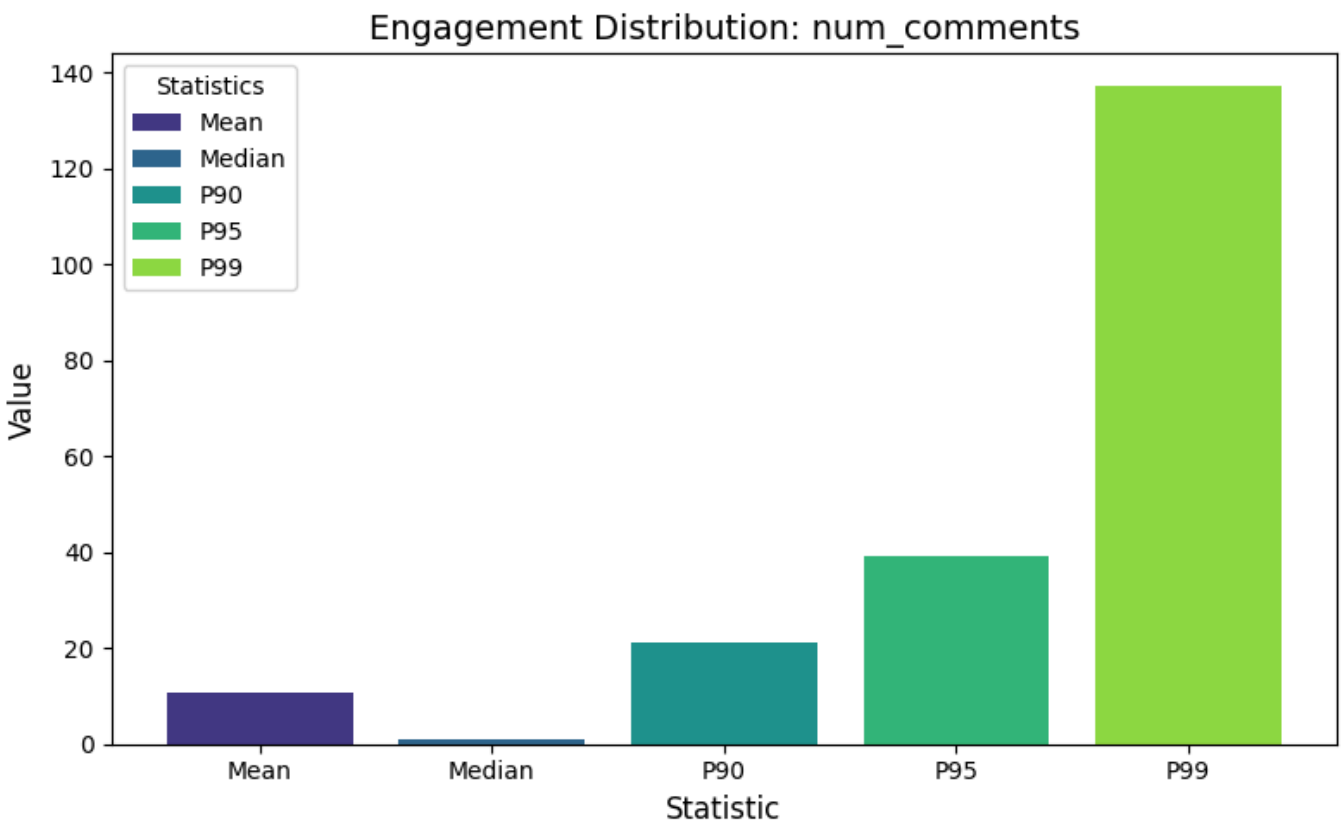
Sentiment Polarity Stats



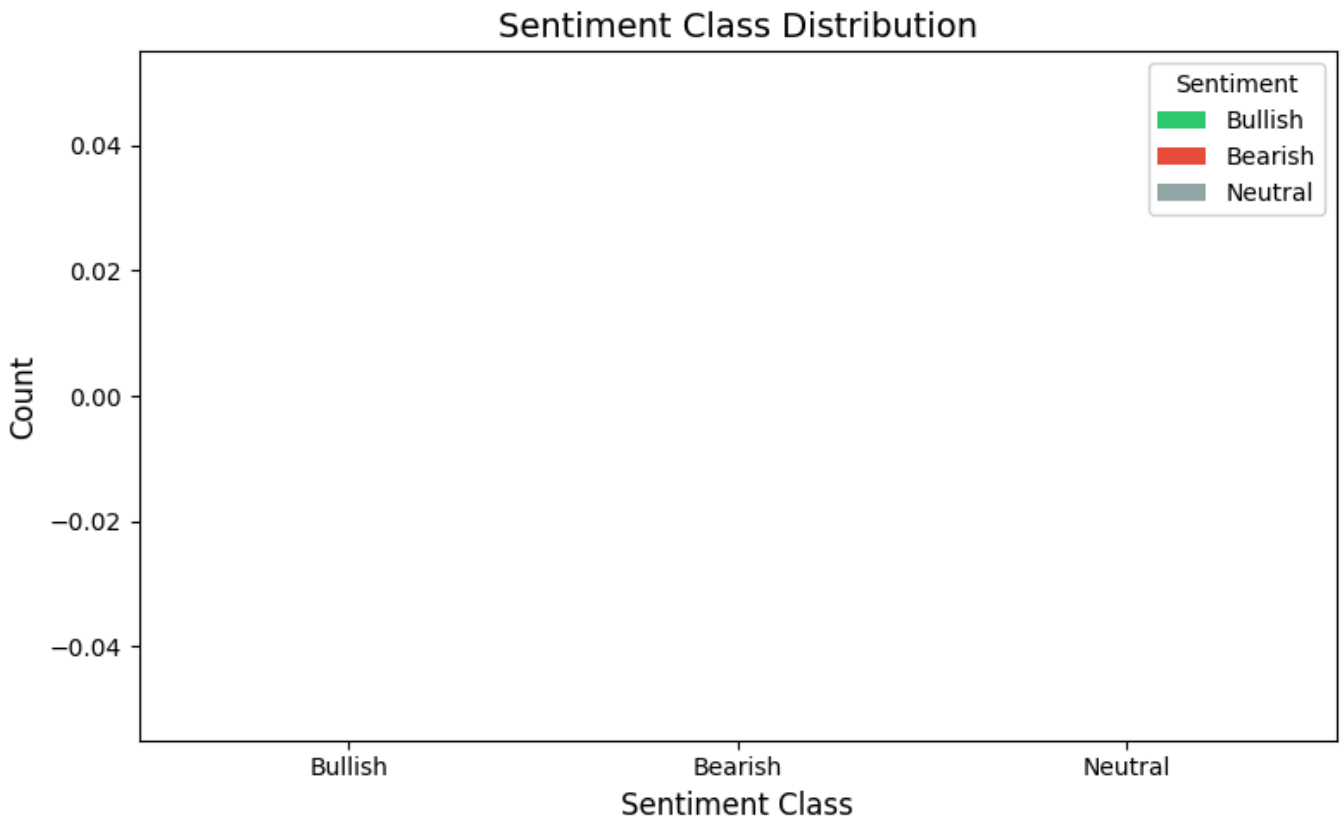
Engagement Distribution Score



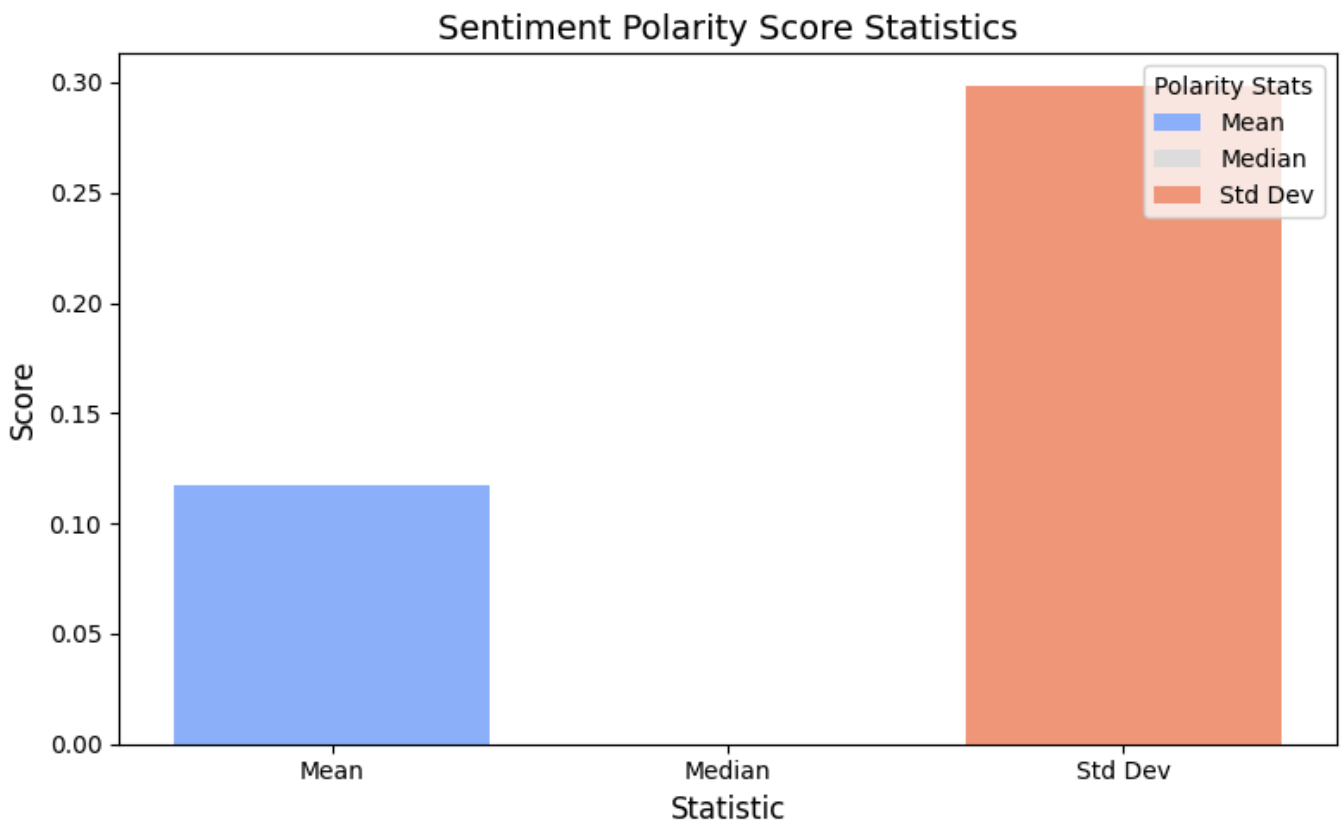
Engagement Distribution Num Comments



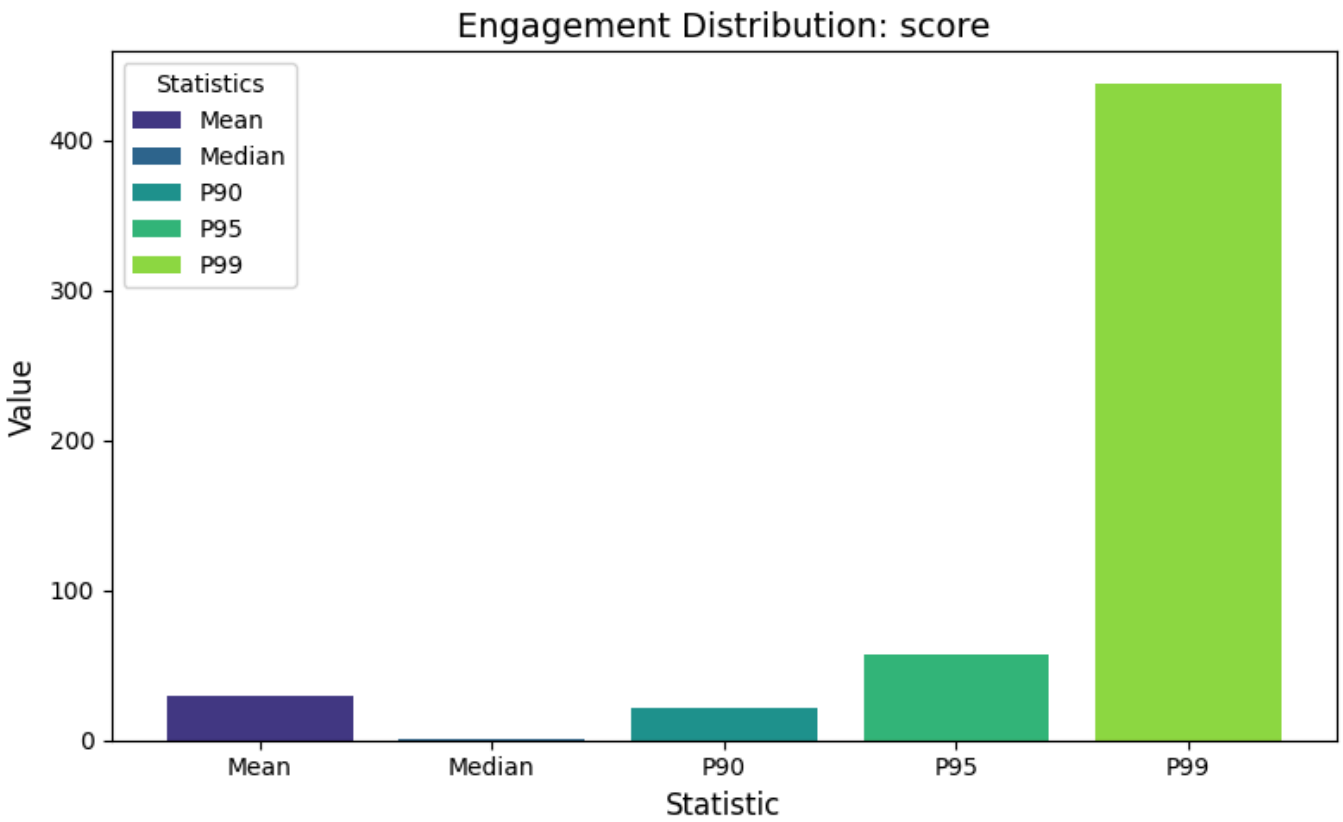
Sentiment Class Distribution



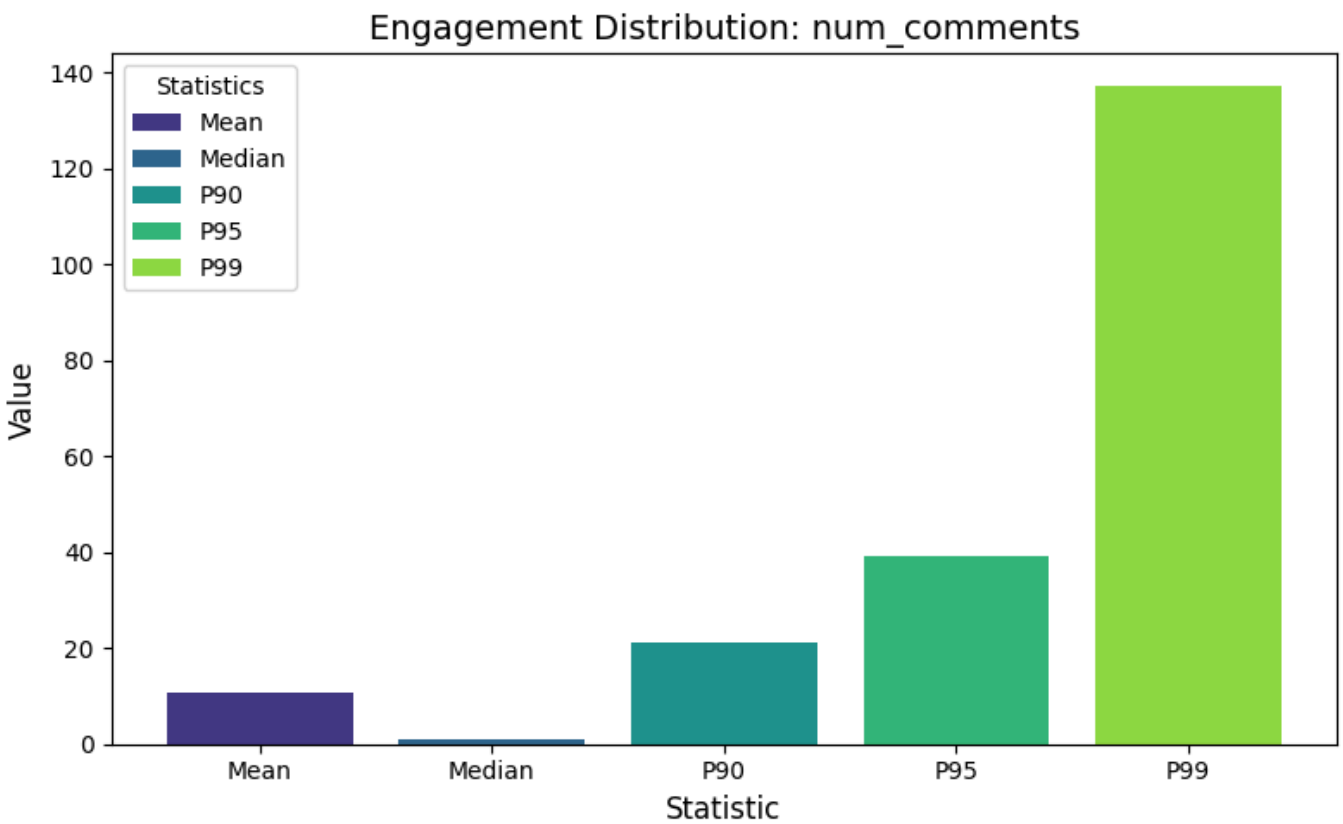
Sentiment Polarity Stats



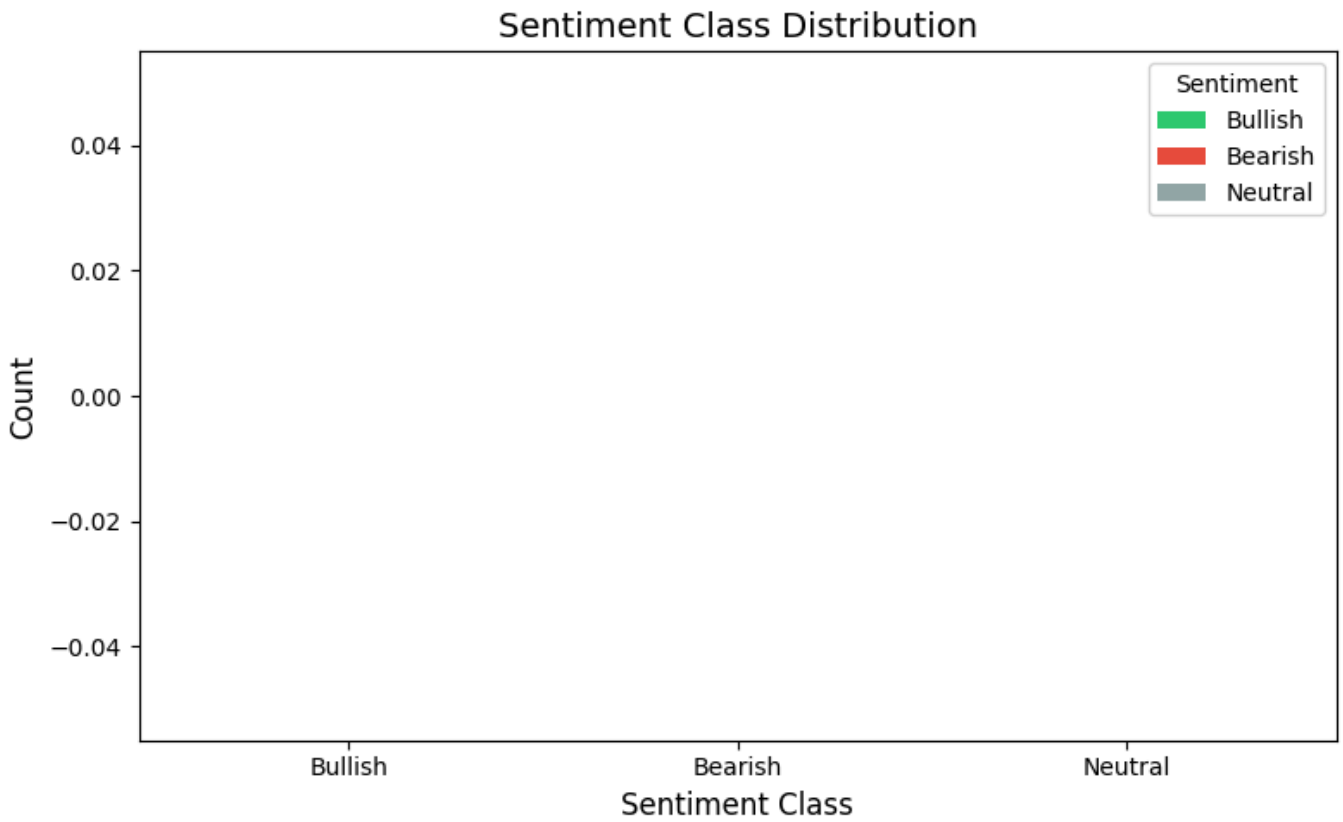
Engagement Distribution Score



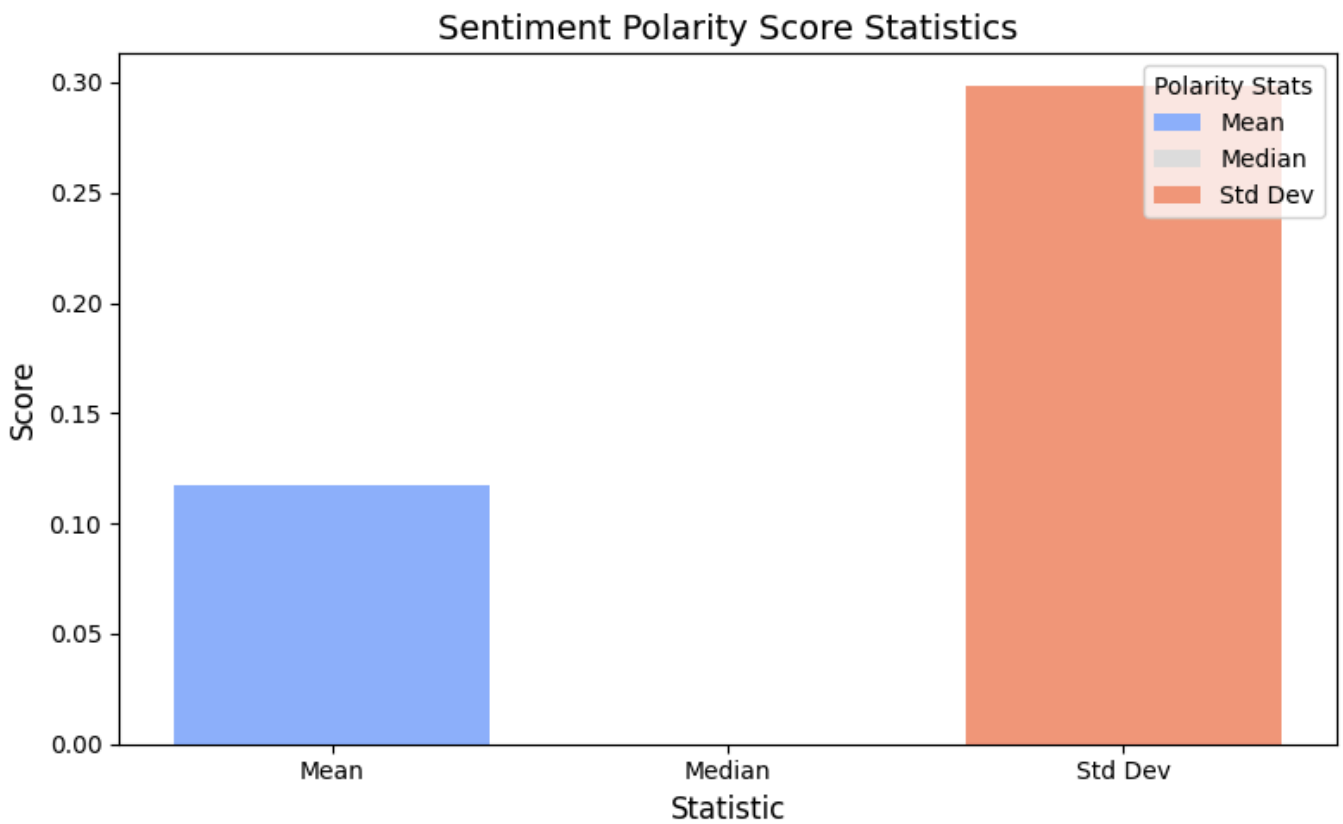
Engagement Distribution Num Comments



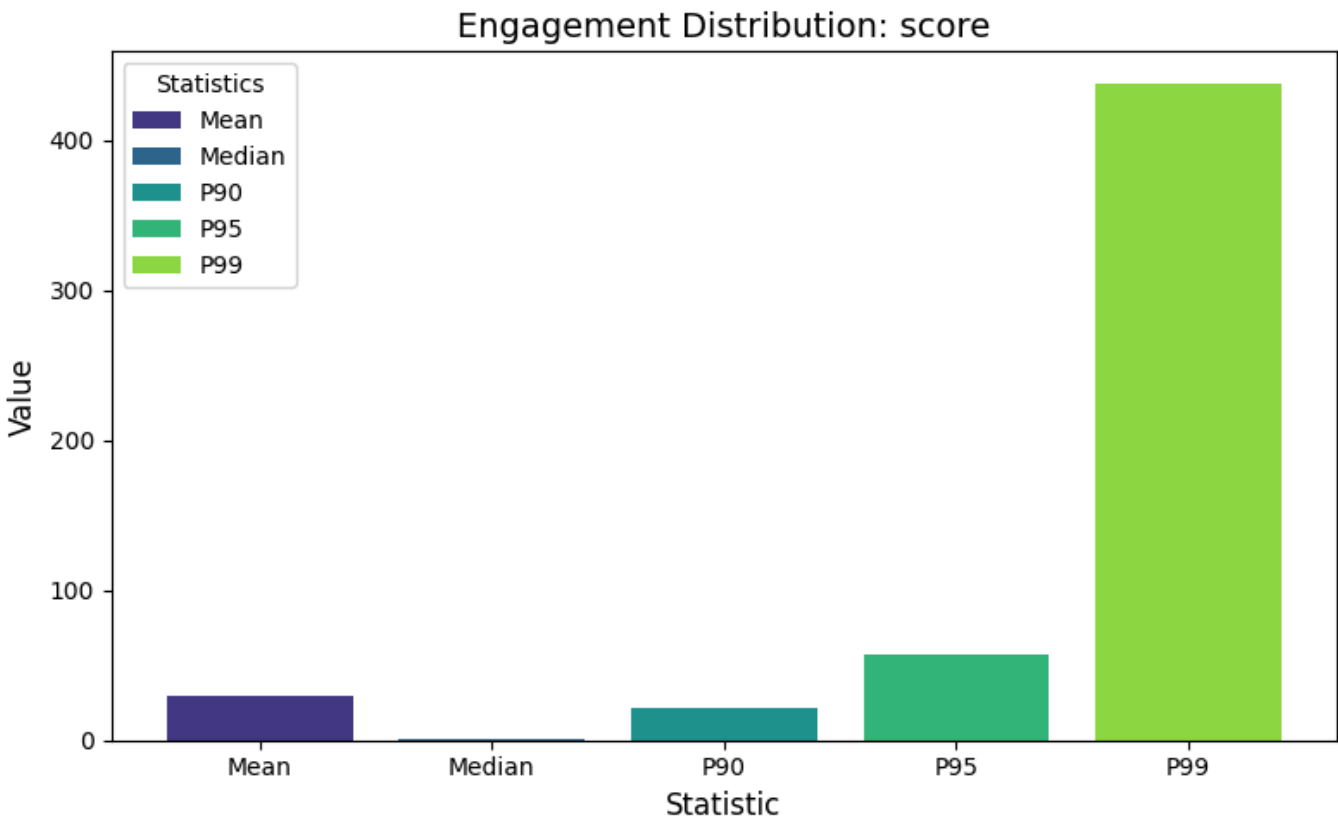
Sentiment Class Distribution



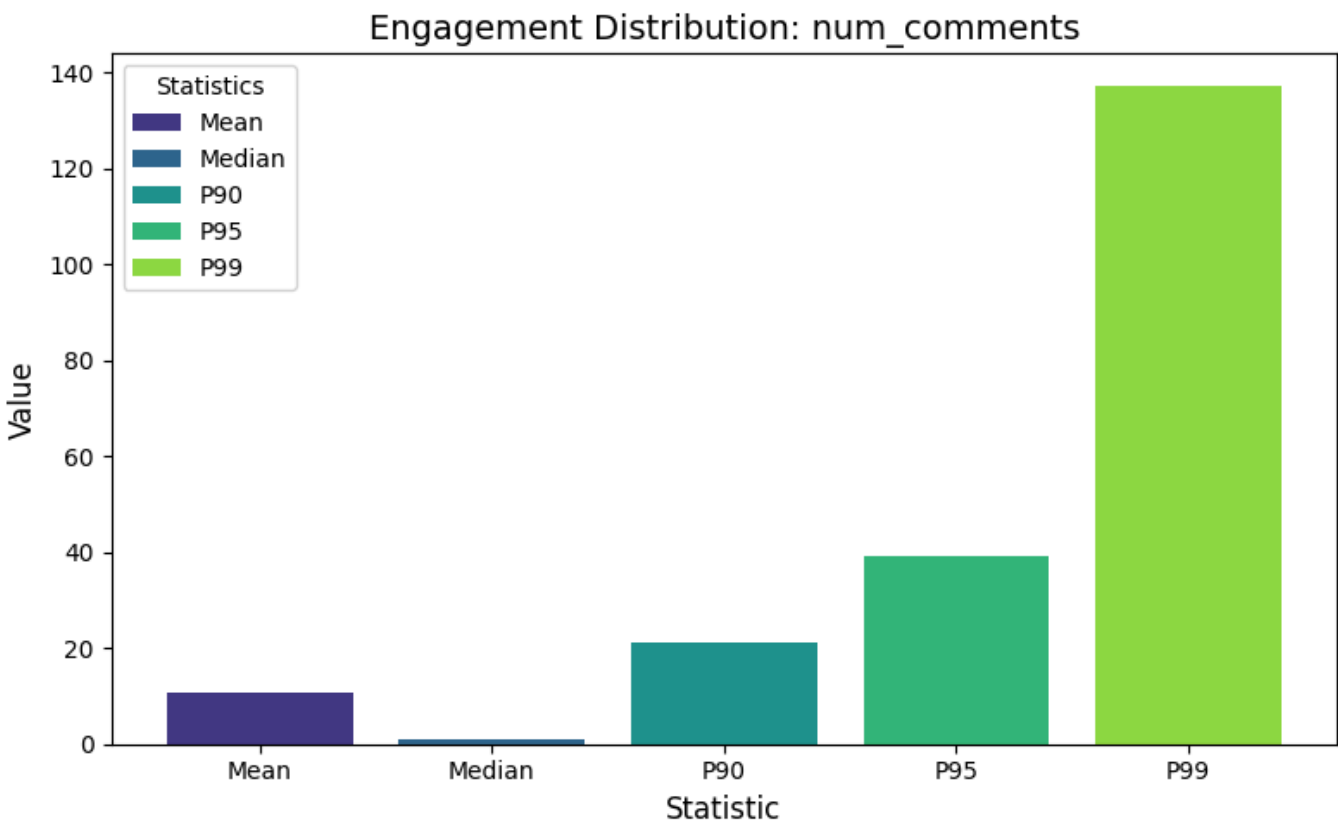
Sentiment Polarity Stats



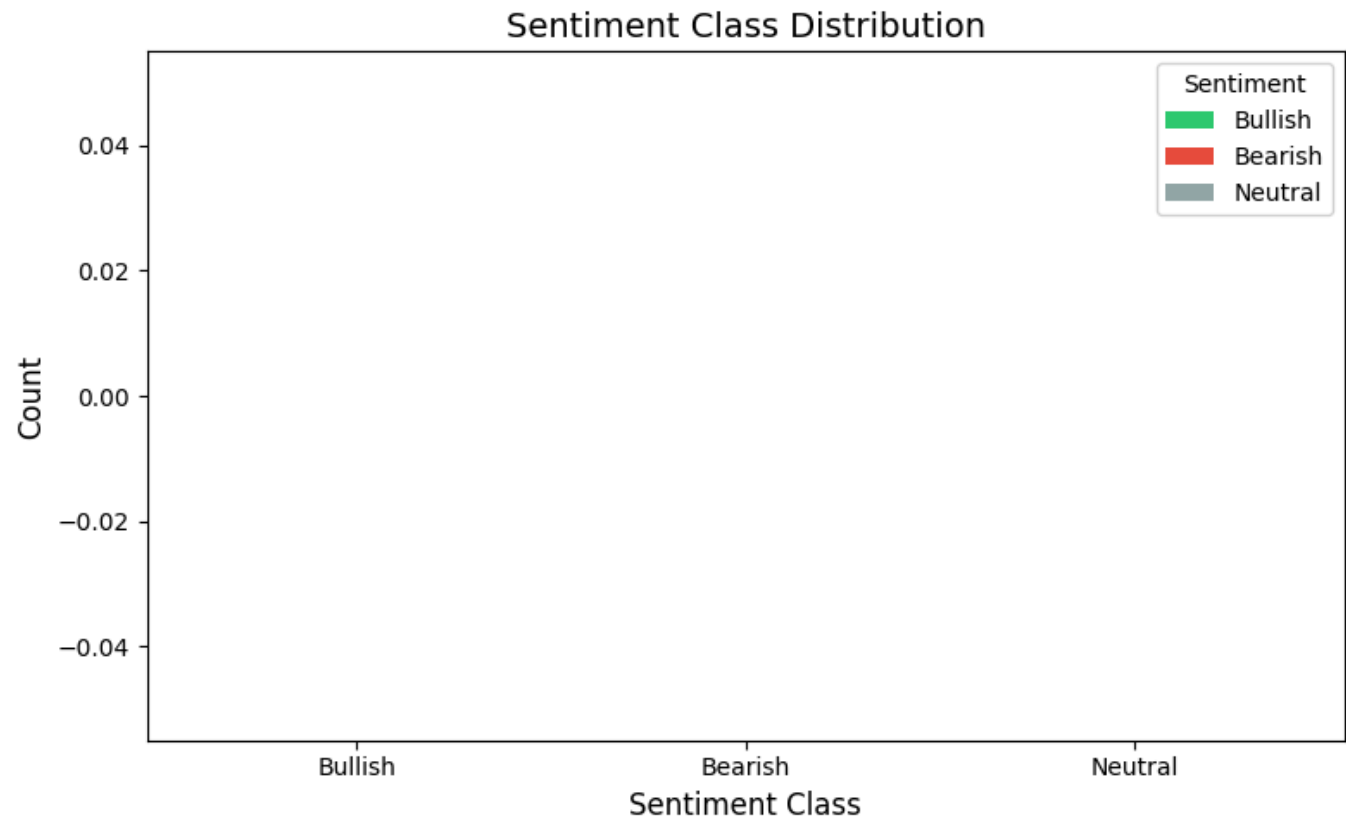
Engagement Distribution Score



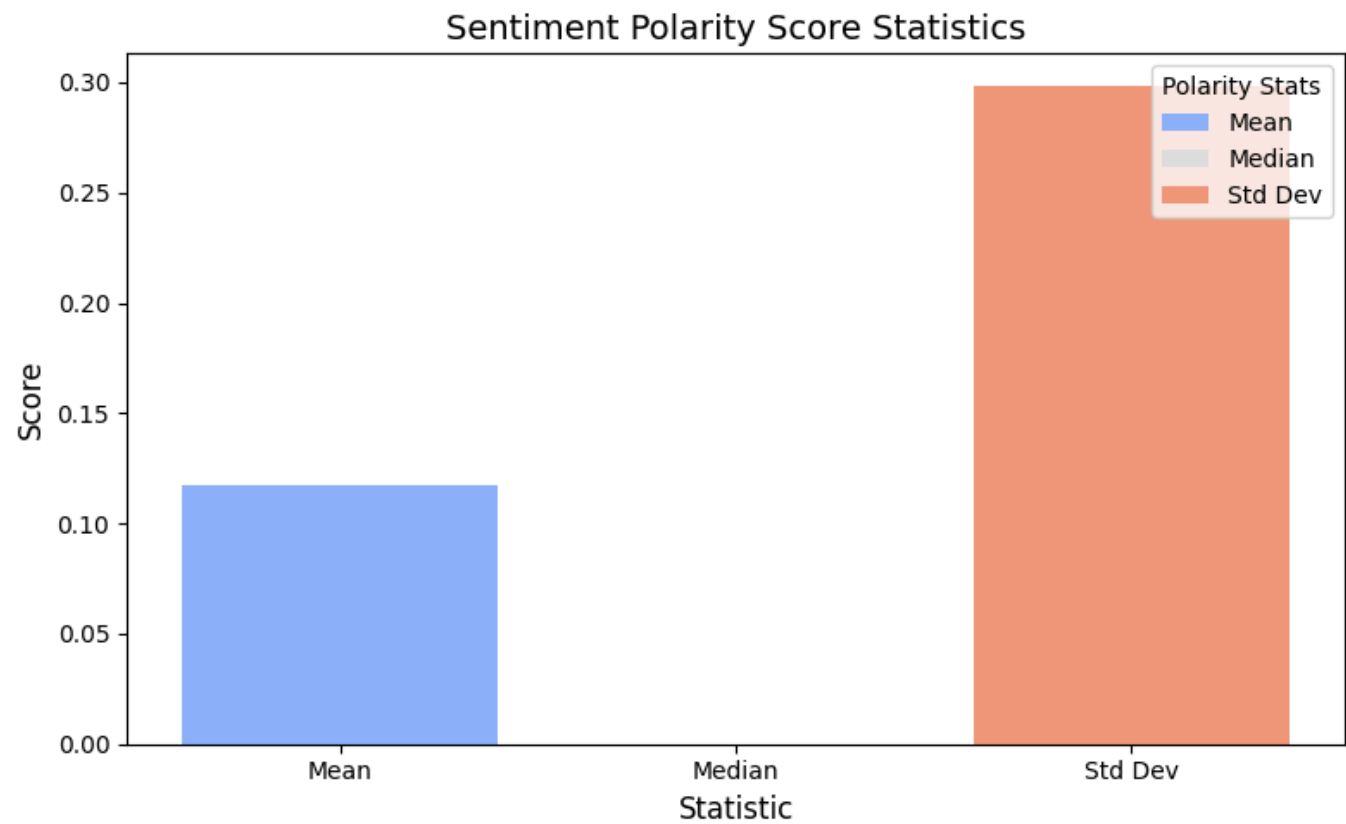
Engagement Distribution Num Comments



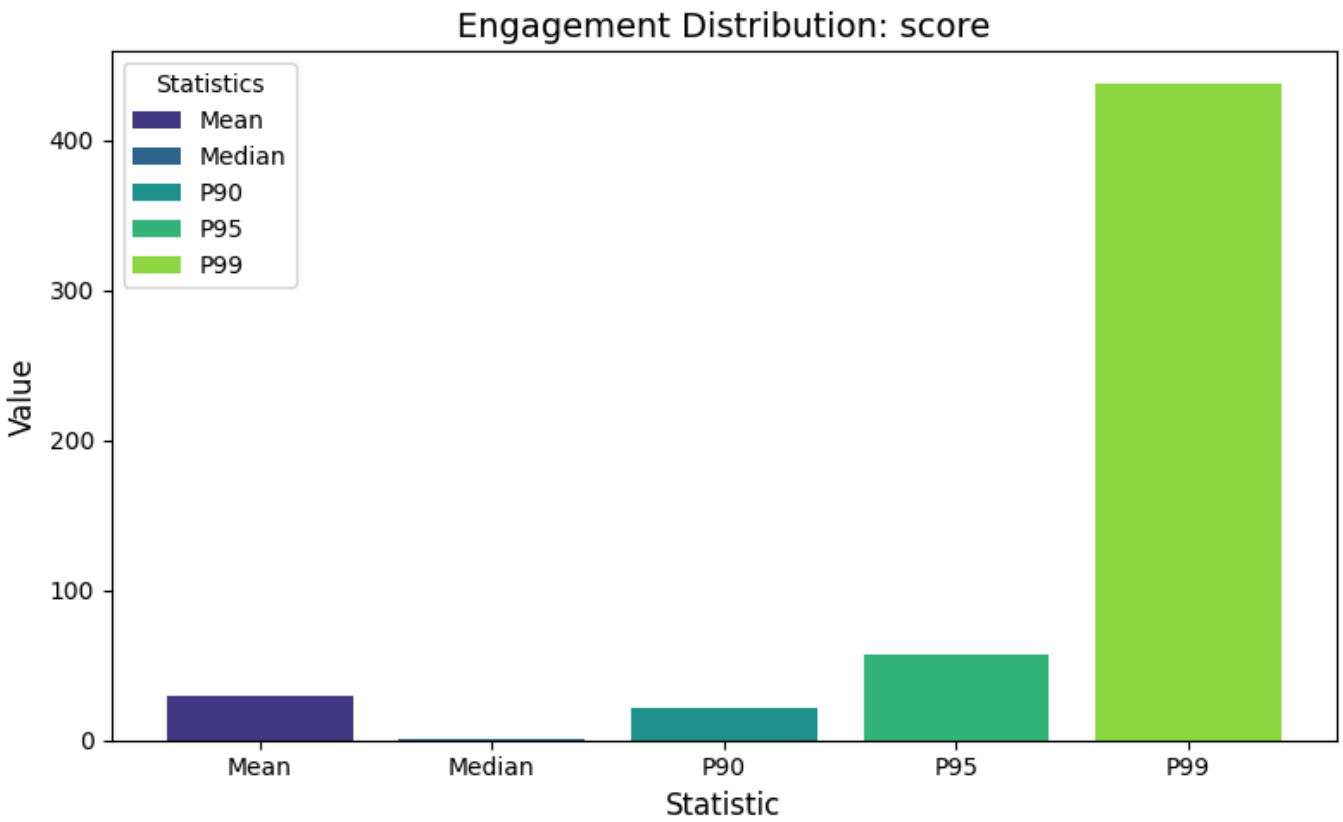
Sentiment Class Distribution



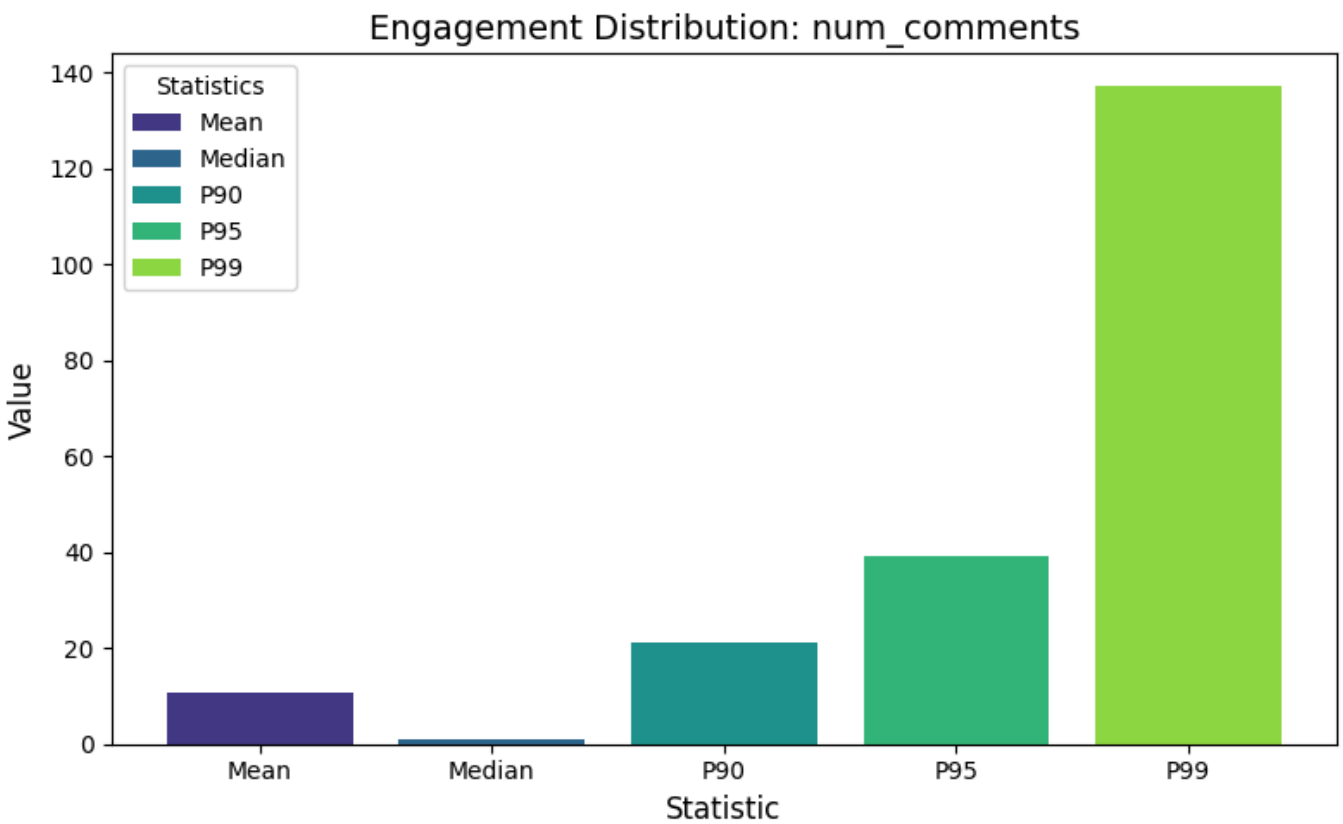
Sentiment Polarity Stats



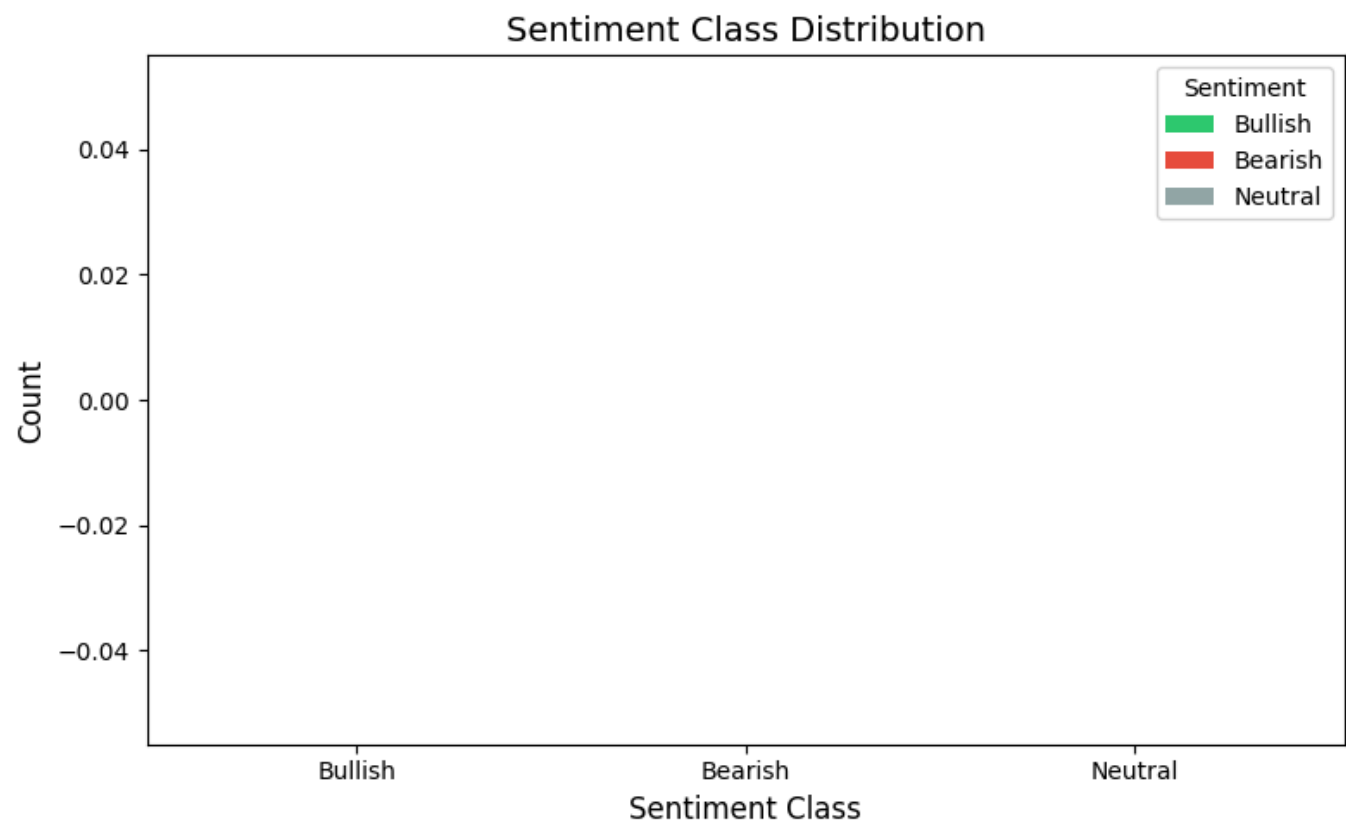
Engagement Distribution Score



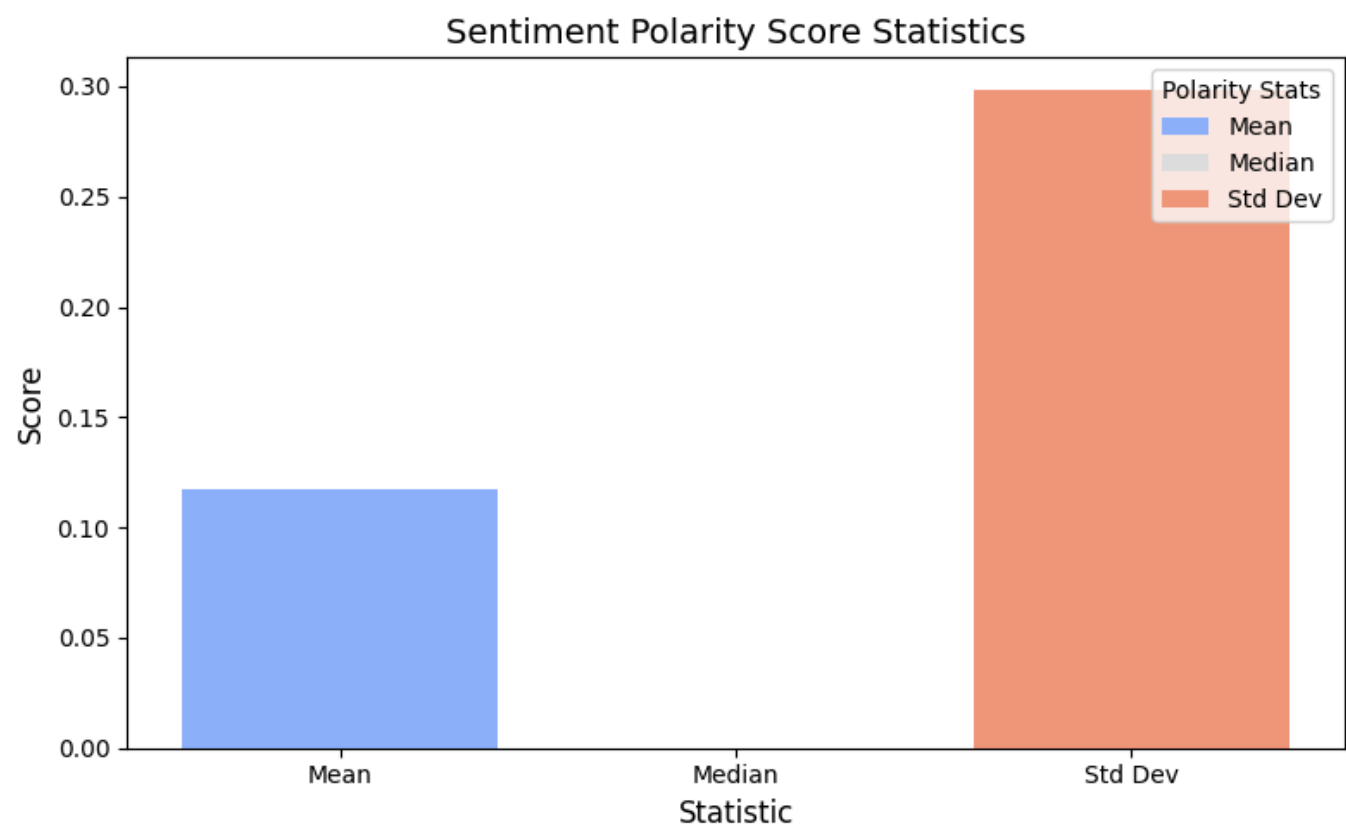
Engagement Distribution Num Comments



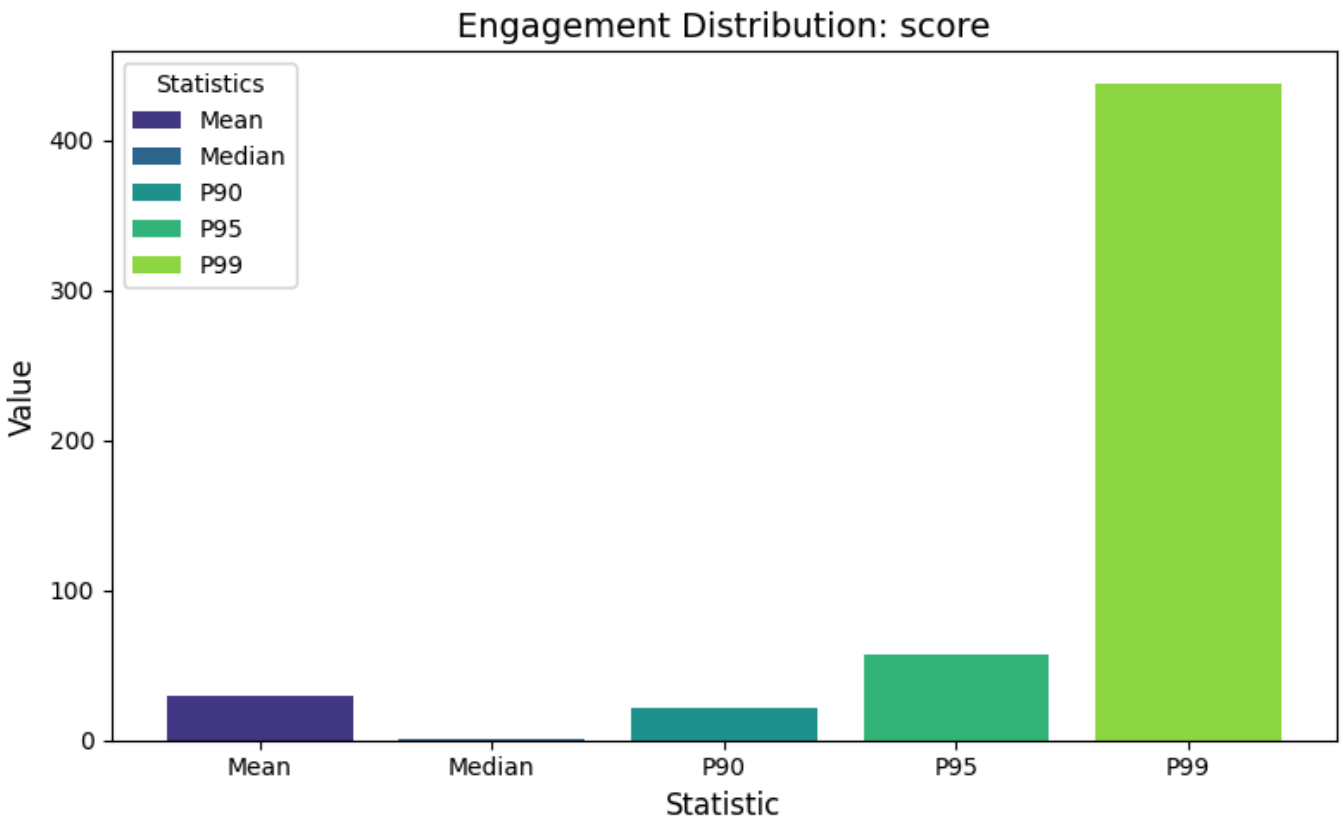
Sentiment Class Distribution



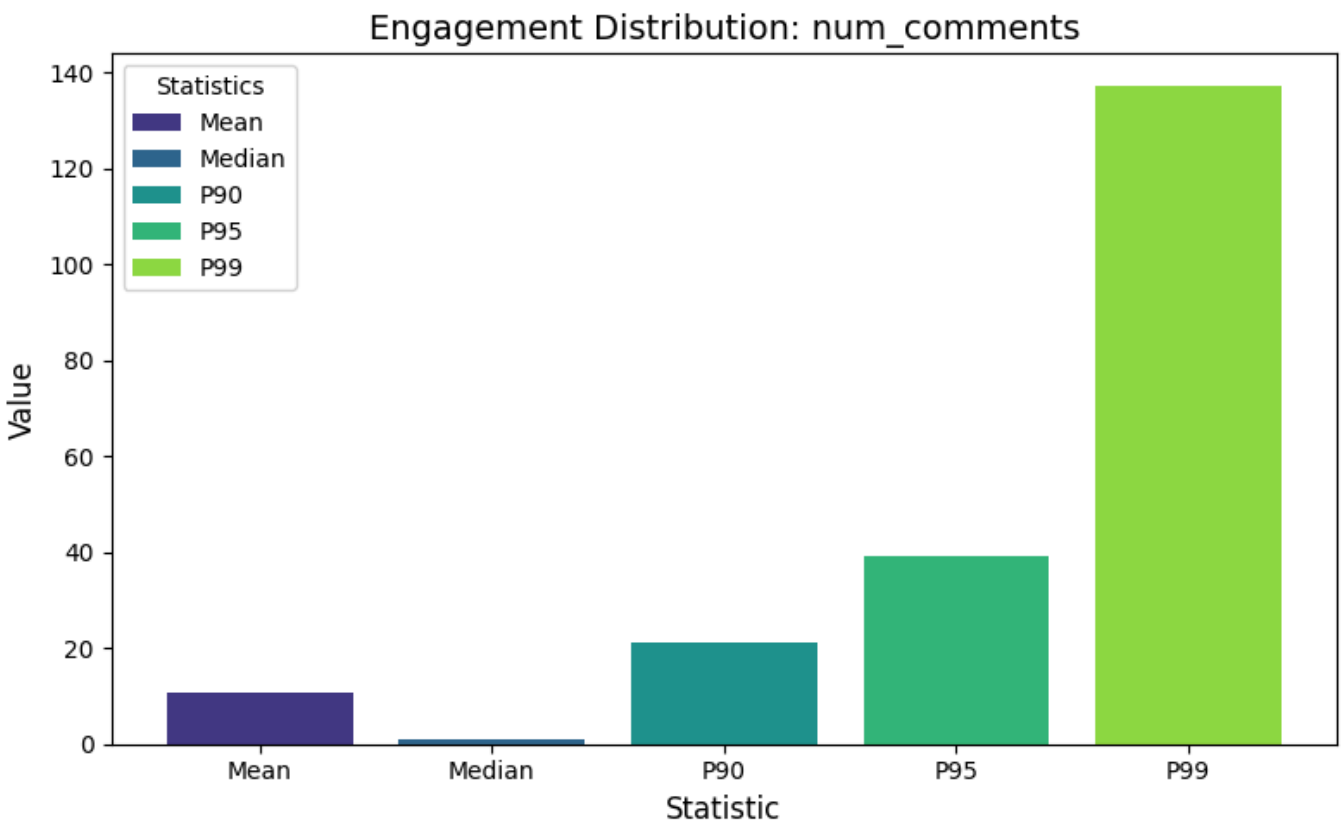
Sentiment Polarity Stats



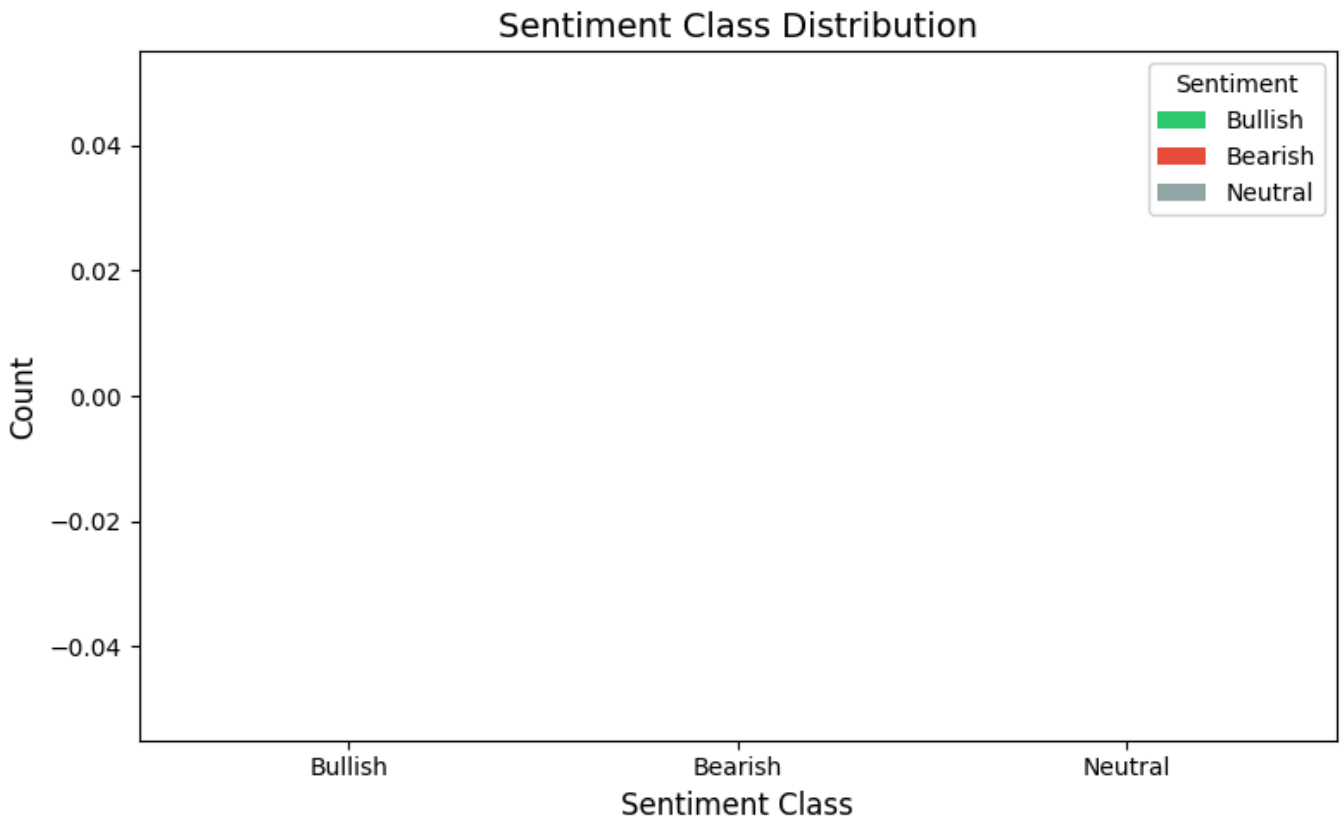
Engagement Distribution Score



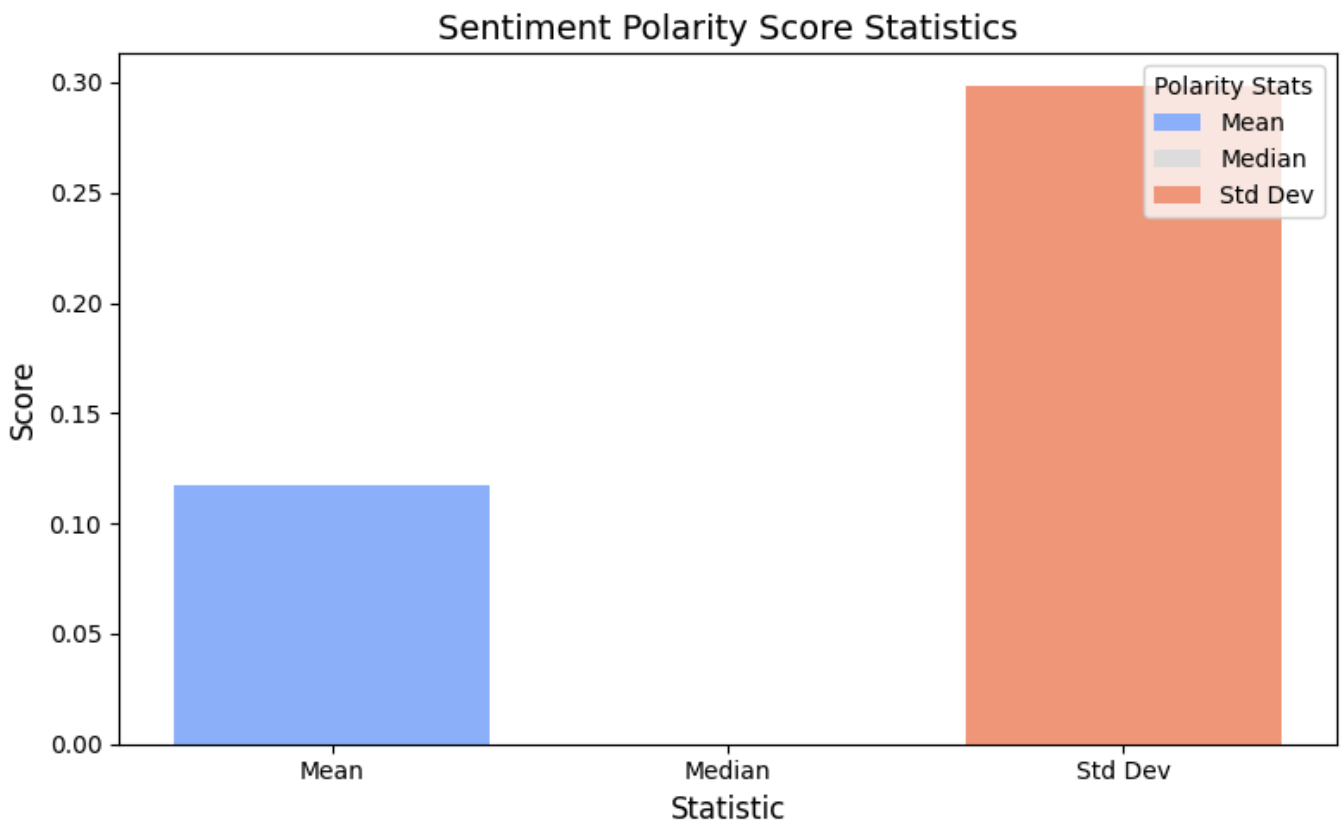
Engagement Distribution Num Comments



Sentiment Class Distribution



Sentiment Polarity Stats



Pipeline Performance

Stage Timings

Stage	Duration (s)
Dataset Discovery	18.97
API Feasibility	0.00
Dataset Quality	431.72
Surge Analysis	50.69
Visualization	4.53
Total	505.92

Per-Dataset Processing Time

Dataset	Duration (s)
StockMarket_subreddit.csv	24.57
submissions_reddit_1.csv	274.82
submissions_reddit_2.csv	25.29
submissions_reddit_3.csv	16.13
submissions_reddit_4.csv	50.81
submissions_reddit_5.csv	19.90
train_data.csv	0.12

Final Recommendation

Recommended Path: Public Dataset (submissions_reddit_4.csv)

Recommend public dataset 'submissions_reddit_4.csv' as the best data path. Key strengths: good data completeness, sufficient record volume, adequate temporal coverage. Public datasets offer immediate availability without collection delays or API costs.

Ranked Options

1. **submissions_reddit_4.csv** (overall score: 1.000)
- High data completeness with few missing values
 - Large dataset suitable for model training
 - Good temporal coverage with minimal gaps

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	1.00	25%	Excellent - very few missing values
Volume	1.00	20%	Excellent - 100k+ records

Criterion	Score	Weight	Comment
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	1.00	20%	No risks identified

2. **reddit API** (overall score: 1.000)

- Low cost for data collection
- Fast collection time
- Supports surge label construction
- Some fields require paid access: full historical archive, real-time streaming

Scoring Breakdown:

Criterion	Score	Weight	Comment
Cost	1.00	25%	Free access
Collection Time	1.00	20%	Very fast (≤ 1 hour)
Feasibility	1.00	25%	Supports surge label construction
Historical Access	1.00	15%	Historical data access available
Field Availability	1.00	15%	Rich field set (10+ fields)

3. **submissions_reddit_5.csv** (overall score: 0.960)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	1.00	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	1.00	20%	No risks identified

4. **submissions_reddit_1.csv** (overall score: 0.959)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps

- 1 risk(s) identified: High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	1.00	20%	Excellent - 100k+ records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

5. **StockMarket_subreddit.csv** (overall score: 0.930)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: Duplicate rows detected: 1103 duplicates (1.5%), which may inflate engagement statistics.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	1.00	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

6. **submissions_reddit_2.csv** (overall score: 0.919)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records

Criterion	Score	Weight	Comment
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

7. **submissions_reddit_3.csv** (overall score: 0.919)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

8. **twitter API** (overall score: 0.770)

- Fast collection time
- Supports surge label construction
- No historical data access - requires prospective collection
- Some fields require paid access: impression_count, full archive search, quote_count

Scoring Breakdown:

Criterion	Score	Weight	Comment
Cost	0.50	25%	Moderate cost (\$50-\$200)
Collection Time	1.00	20%	Very fast (≤ 1 hour)
Feasibility	1.00	25%	Supports surge label construction
Historical Access	0.30	15%	No historical access - prospective only
Field Availability	1.00	15%	Rich field set (10+ fields)

9. **train_data.csv** (overall score: 0.750)

- High data completeness with few missing values
- Large dataset suitable for model training

- Good temporal coverage with minimal gaps
- 3 risk(s) identified: Duplicate rows detected: 16970 duplicates (99.9%), which may inflate engagement statistics., No date column 'date' found: temporal analysis and 24-hour surge windowing cannot be performed., No ticker column 'ticker' found and ticker extraction from text yielded no results: per-ticker engagement normalization cannot be performed.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	1.00	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	0.20	15%	Very limited - fewer than 3 tickers
Risk	0.55	20%	Moderate risks (3-4 issues)